Ali Nikkhah

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Summary

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation. — Variant: Academic / Research

Technical Skills

Languages: Python, C++, Go, SQL, R, Bash, TypeScript
ML/Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper
Agentic: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE
LLM Serving/GPU: vLLM, TGI, AWQ, GPTQ, KV-cache
Data/MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray

Professional Experience

Iran Credit Scoring Bureau — Tehran, Iran
ML Engineer — Self-Hosted Graph RAG Assistant (Concurrent) — *Apr 2026 == Present*
- Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.
- Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.
- Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.
- Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.
- *Stack:* Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain, FAISS, vLLM
- *Most recent Iran position; overlaps Sharif M.Sc. AI (Sep 2025) — full-time industry alongside full-time graduate study*

Sharif University — Robust & Interpretable ML Lab — Tehran, Iran
Research Collaborator — Explainable & Robust AI (Concurrent) — *Jul 2025 == Sep 2025*
- Benchmarked adversarial + natural shift suites; notebooks surfacing Captum/SHAP attributions, saliency, counterfactuals.
- Documented deploy best-practices for regulated industries with faculty.
- *Stack:* PyTorch, MLflow, Captum, SHAP
- *Summer research sprint concurrent with industry roles*

Trinity College Dublin — EmoDub Project — Dublin, Ireland
Machine Learning Researcher — Emotion-Aware Speech Translation (Concurrent) — *May 2025 == Nov 2025*
- Trained SER front-end (transformer + wav2vec2.0 + Whisper) for affect logits; cross-modal fusion transformer for emotion-consistent decoding.
- Curated emotion-annotated corpora (EmotionNet/EmoDB) with balanced cultural representation; evaluated BLEU, Emotion F1, MOS.
- Collab. with Prof. Siobhán Clarke; reproducible codebase https://github.com/AliNikkhah2001/emodub-emotion-translation.
- *Stack:* PyTorch, TensorFlow, Hugging Face, librosa, wav2vec2.0, Whisper, EmotionNet, EmoDB
- *Part-time remote research concurrent with Digikala + Turquoise + UBC*

University of British Columbia — Remote · Vancouver, Cana
Research Assistant — Vision-Language Medical Imaging (Ultrasound Report Generation) (Concurrent) — *Feb*
- Designed ViT/BLIP encoders + T5 generator with contrastive/self-supervised alignment; curated DICOM, de-identified QA corpora.
- Deployed attention/saliency overlays for interpretability; benchmarked BLEU/ROUGE-L + clinical efficacy with radiologists; SOTA on in-house split.
- Preprint in prep: Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning (2025).
- *Stack:* PyTorch, ViT, BLIP, T5, LLaVA, CLIP, OpenCV, MIMIC-CXR
- *Long-running part-time research; overlapped industry roles but capped at 10h/wk — primary commitment was Digikala/Turquoise*

Sharif University of Technology Tehran, Iran
Teaching Assistant — NLP, Generative Models, Deep Learning (Concurrent) *Aug 2023 == Jan 2024*
- Facilitated seminars, office hours (30-80 students), problem-set design/grading, exam design for DL/NLP.
- Developed lab material for Deep Learning, Analog Electronics, foundational ML; mentored undergrad capstones.
- *Stack:* PyTorch, TensorFlow, NumPy, scikit-learn
- *Graduate TA alongside L3S research — teaching load 6-8h/wk*

L3S Research Center — Leibniz University Hannover Hannover, Germany
Research Assistant — Video Motion Classification (Concurrent) *Apr 2023 == Feb 2024*
- Engineered optical-flow-centric datasets with temporal sampling/augmentation; improved domain generalization across lighting/texture/camera shift.
- Benchmarked 3D CNN + transformer hybrids vs baselines; analyzed attention maps; reduced overfitting, SOTA accuracy.
- *Stack:* PyTorch Lightning, OpenCV, PyAV, TensorBoard, scikit-learn, Multimodal Models
- *Part-time RA during B.Sc. final year — no conflict*

Sharif University of Technology Tehran, Iran
Teaching Assistant — History (8 courses, 2021-2023) *Jan 2021 == Aug 2023*
- Graded problem sets, ran tutorials, held office hours; supported 30-80 students per term.
- *Stack:* PyTorch, TensorFlow, NumPy

Selected Projects & Research

Trinity College Dublin · EmoDub May 2025 – Nov 2025
Emotion-aware voice translation
- SER front-end + wav2vec2.0 bullets (see experiences)

University of British Columbia Feb 2024 – Nov 2025
Ultrasound report generation

L3S Research Center Apr 2023 – Feb 2024
Texture-free motion intelligence

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning — *arXiv preprint — in preparation*, 2025

Education

Sharif University of Technology Tehran, Iran
M.Sc. Artificial Intelligence Computer Engineering (full-time, ongoing) *Sept 2025 – Present*

Sharif University of Technology Tehran, Iran
B.Sc. Electrical Engineering GPA: 3.65 *Sept 2020 – Sept 2025*